Banajit Barman

PhD Researcher in Experimental High-Energy Physics

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ORCID Inspire HEP GitHub Google Scholar Website

Education

- **PhD in Experimental High-Energy Physics** *Feb 2022 – Oct 2026*
  Gauhati University, India
  - Member of the Collaborator in the ALICE experiment at CERN, Switzerland.
  - Supervisor: *Prof. Buddhadeb Bhattacharjee.*
  - Thesis: *Studies on light flavored particle production in RUN3 pp collisions using ALICE-TOF detector and MC data.*
- **Master of Science (MSc) in Physics** *2019 – 2021*
  Gauhati University, India
  - Specialization: Nuclear Physics and Electronics.
- **Bachelor of Science (BSc) in Physics** *2016 – 2019*
  Nalbari College, under Gauhati University, India.

Research Interest

- Relativistic hadronic collisions experiments and phenomenology.

As an active researcher within the ALICE Collaboration, my primary focus lies in unraveling the collective dynamics of the Quantum Chromodynamics (QCD) medium, currently driving my experimental analysis using the newly collected LHC Run 3 data. My work bridges experimental data analysis and phenomenological modeling, specifically targeting the emergence of medium-like behaviors in small and light-ion collision systems at the LHC energies. Alongside my experimental work, I use the PYTHIA 8 event generator to understand how mechanisms such as multi-parton interactions (MPI) and Color Reconnection (CR) can reproduce features that resemble QGP-like behavior without invoking a thermalized medium. Furthermore, my research extends into baryon number transport and heavy-flavor hadronization.

ALICE Paper/ Analysis Note/ Activities

1. **Production of $\pi^{\pm}$, $K^{\pm}$, and p ($\bar{p}$) in pp collisions at $\sqrt{s} = 13.6$ TeV**, ALICE Collaboration.
   Publication page (internal) (In CERN Review) **(One of Paper Committee (PC) members and PC chair)**
   Analysis note (internal)
2. **Production and Nuclear Modification factor of light flavoured particles in OO collision at $\sqrt{s_{NN}} = 5.36$ TeV**
   Analysis note (internal) **(One of Principal Analyzers)**
3. ALICE Collaboration (S. Acharya *et al.*). **"Systematic study of flow vector fluctuations in $\sqrt{s_{\rm NN}} = 5.02$ TeV Pb–Pb collisions."** *Physical Review C* **109**, 065202 (2024).**(Contributed in Institutional Review)**
4. ALICE Collaboration (S. Acharya *et al.*). **"Measurement of the Λ hyperon lifetime."** *Physical Review D* **108**, 032009 (2023). **(Contributed in Institutional Review)**

ALICE Internal Talk

1. **Particle identification in O2Physics**. O2 Analysis Tutorial 3.0, CERN, Switzerland. *Nov 2023*
   [Contribution Link] (Talk)

ALICE Data-Taking Operations/ Service Work

- **Shift Leader:** Managed 5 shifts (2024–2025) at the ALICE Run Control (ARC). Responsibilities included pre-shift system verification, directing run configurations based on beam conditions, overseeing the crew, and managing detector resource locks to ensure high-quality data acquisition during LHC Run 3.
  - *2025:* 10–15 June, 18–23 June.
  - *2024:* 27 Aug–01 Sept, 10–15 Sept, 20–25 Sept.
- **ECS Operator:** Executed 2 shifts (2022) at the ARC. Responsibilities included managing the experimental state machine, monitoring critical sub-detector parameters (HV/LV, gas, cooling), and executing safety interlocks to ensure hardware protection and stable data acquisition.
  - *2022:* 06–11 Sept, 14–19 Sept.
- **Service Work (Nov 2023 – Dec 2024):** Performed data quality monitoring for the ALICE Data Preparation Group (DPG). Key tasks included validating Particle Identification (PID) performance for the TPC and TOF detectors using AO2D data.

Phenomenological Studies

1. Tonmoy Sharma, **Banajit Barman**, and Buddhadeb Bhattacharjee. **"Production of heavy flavored particles in $p + p$ collisions: A unified model description for ALICE and LHCb Measurements."** *Physical Review C* **113**, 034918 (2026).
2. **Banajit Barman**, Nur Hussain, and Buddhadeb Bhattacharjee. **"Baryon number transportation over full rapidity space in pp collisions at energies available at the CERN Large Hadron Collider."** *Physical Review C* **111**, 014905 (2025).
3. Pranjal Sarma, **Banajit Barman**, and Buddhadeb Bhattacharjee. **"Effect of color reconnection on multiplicity-dependent charged particle production in PYTHIA-generated pp collisions at the LHC energies."** *The European Physical Journal A* **59**(4), 76 (2023).

Conference Proceedings

1. **Banajit Barman** and Buddhadeb Bhattacharjee. **"On transportation of baryon number in pp collisions at $\sqrt{s} = 0.9$ and 7 TeV."** *Journal of Subatomic Particles and Cosmology* **4**, Article 100121 (2025).
2. D. Choudhury, **B. Barman**, B. Bhattacharjee, et al. (ALICE Collaboration). **"Light-flavoured particle production in pp collisions at 13.6 TeV with ALICE at the LHC."** *DAE Symposium on Nuclear Physics* **67**, 993–994 (2024).

Oral / Poster Presentations

1. **Production of light-flavoured particles in ALICE Run 3 data for pp collisions**. 10th Asian Triangle Heavy-Ion Conference (ATHIC 2025), Berhampur, India. *Jan 2025*
   [Contribution Link] (Oral)
2. **On Transportation of Baryon Number in pp Collision at $\sqrt{s} = 0.9$ and 7 TeV**. 10th Asian Triangle Heavy-Ion Conference (ATHIC 2025), Berhampur, India. *Jan 2025*
   [Contribution Link] (Oral)
3. **Multiplicity-dependent π, K, p production in pp collisions at 13.6 TeV using ALICE TPC and TOF detectors**. Quark Matter 2023, Houston, TX, USA. *Sep 2023*
   [Contribution Link] (Poster)

4. **Particle identification in LHC Run 3 pp collisions at $\sqrt{s}$ = 13.6 TeV using the ALICE TPC and TOF detectors**. ICPAQGP-2023, Puri, Odisha, India. *Feb 2023*
[Contribution Link] (Oral)

Licenses & Certifications

Artificial Intelligence Fundamentals Issued by: *IBM*, Jul 2026

Python for Data Science Issued by: *IBM*, Jul 2026

Achievements/Awards

- Awardee of Breakthrough Prize in Fundamental Physics (2025) – ALICE Collaboration
- Junior Research Fellowship (JRF), India (Jun, 2022 – Oct, 2026)
- Graduate Aptitude Test in Engineering (GATE) – Physics (2022)
- Ishan Uday Scholarship (2016-2019)

Skills

- **Programming:** Proficient in C++, Python, Julia and Parallel computing with experience in data analysis.
- **Data Analysis Tools:** Expertise in O2 framework (based on C++17) of ALICE experiment and ROOT by CERN.
- **Collaboration:** Teamwork experience as a CERN ALICE collaboration member.
- **Languages:** Assamese (Native), English (Fluent), Hindi (Fluent)